

Tinjau

Verified background checks for AI agents: facts about ERC-8004 agents and their reviewers, proven from Ethereum into Creditcoin through the Attestcoin Protocol.

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Abstract

Over 19,000 AI agents are registered on ERC-8004 on Ethereum mainnet and already hire and pay each other. Nothing a contract can check says which of them deserve the money. Read raw, the registry misleads: most rated agents carry a single reviewer, one wallet wrote hundreds of reviews, and most recent registrations come from owners operating at clone scale. The standard's own answer is to aggregate off-chain and trust the aggregator.

Tinjau is a bureau in Creditcoin's original sense. It records and it does not judge. Its Creditcoin contract `GroundedFacts` admits data **only** through Attestcoin proofs of Ethereum transactions, verified by the `BlockProver` precompile inside the transaction that writes them. It stores facts, never opinions, and it emits no score. A consumer passes its own thresholds and receives admitted evidence together with the gaps that are known to exist, and anyone can replay every proof from chain data alone and arrive at the same numbers.

1. The problem, measured

These figures were measured directly against Ethereum mainnet on 29 August 2026 by decoding registry events over an RPC connection. They are not estimates.

Observation	Measurement	Method
Reviewer concentration	Of the last 600 reviews: 367 agents, 106 unique reviewers; 346 of 367 agents have exactly one reviewer; one externally owned account wrote 225 reviews for 195 agents	decoded <code>NewFeedback</code> events
Conflicted reviewers	16 of 105 reviewers own agents themselves, and wrote 59% of all feedback	registry cross-reference
Clone-scale registration	Last 60 days: 14,771 registrations, 83% from owners holding ten or more agents; 8,136 agents minted inside multi- <code>Registered</code> transactions	<code>eth_getLogs</code> , blocks 25,429,616 to 25,861,616
Unverifiable payment claims	0 of 12 sampled "paid on Ethereum" claims inside review documents exist on the chain they name	prover lookup per claimed hash

Independent work reaches the same conclusion from another direction: arXiv 2606.26028 finds 73.5% coordinated Sybil reviewers on Ethereum. A rating in this registry is therefore not evidence of anything until someone checks who produced it, when they first existed, and whether the agent's own owner controls them.

2. Why the existing answer is not enough

ERC-8004 states that aggregation will happen off-chain. In practice that means an indexer reads the registry, computes a reputation, and signs it. A contract that wants to act on that number must trust the indexer, its RPC provider, and its uptime. The trust simply moves, and it moves to a party with no bond and no penalty for being wrong.

The obstacle is structural rather than one of effort. An EVM contract cannot read its own chain's history, and a contract on a different chain cannot read Ethereum at all. Any on-chain consumer is therefore forced back onto a

signed assertion from somebody.

3. Design principles

1. **Facts, not scores.** The contract computes no weight and issues no verdict. Value comes from refusing to judge, because the party carrying the risk is the one entitled to set the bar.
2. **Only proofs enter.** Every state change passes through `verify` on the BlockProver precompile. There is no admin, no setter, no trusted relayer and no upgrade path.
3. **Omission must never pay.** A reviewer counts as grounded only when every one of their review indices is proven, so withholding evidence can never improve an agent's standing.
4. **Logs only from the official registries.** An event is read only when its emitter matches the ERC-8004 registry configured for that source chain.
5. **No language model in the fact path.** A model reads claims and produces reports; it never writes a fact.
6. **Everything recomputable.** Any number the contract reports can be reproduced off-chain from the same proofs.

4. How a fact enters

A submitter, called a scout, finds an Ethereum transaction that carries evidence, fetches an Attestcoin proof for it, and calls `record`. Inside that call, and before a single byte of the payload is read, the contract does the following.

```
record(Proof[])
  for each proof:
    BlockProver 0x..0FD2  verify(chainKey, height, encodedTx, merkleProof, continuityProof)
                        rejected          -> revert ProofRejected(i)
    BlockProver 0x..0FD2  calculateTxIndex(merkleProof)
                        key = (chainKey, height, txIndex)  duplicate -> skip, never revert
    EvmV1Decoder          decode sender, receipt status, logs
                        status == 0          -> logs discarded, sender activity still counts
                        emitter != registry -> log ignored
    AttestorStash 0x..0FD4 getAttestorsCount(chainKey)  stored alongside the fact
    ChainInfo 0x..0FD3  attested tip, exposed as attestedTip(chainKey)
```

Keying by `(chainKey, height, txIndex)` lets several independent scouts race without double counting, and it orders ownership transfers exactly by height, transaction index and log index, so the newest proven transfer wins regardless of the order in which proofs arrive.

Why this is load bearing. Remove Attestcoin and `record` would have to trust its caller. Tinjau would become one more signed off-chain aggregator, which is the thing it exists against. With it, a Creditcoin contract checks Ethereum history itself and can lock money on the result.

5. What the bureau stores

`facts(chainKey, agentId, minAge, minDepth)` returns thirteen fields. None is a score.

Field	Meaning
<code>breadthRaw</code> / <code>breadthGrounded</code> / <code>breadthIndependent</code>	reviewers seen; those whose every index is proven and who clear the age and activity thresholds; those who additionally own no agents
<code>gapCount</code>	reviewers with an unproven index below their highest proven one
<code>negatives</code>	proven negative reviews, net of proven revocations
<code>cloneDensityLB</code> , <code>registrantSiblings</code> , <code>uriSiblings</code> , <code>sameTxSiblings</code>	agents sharing an owner, a registrant, a document URI, or a single minting transaction
<code>firstRegisteredHeight</code> , <code>coveredThrough</code>	first proven registration height, and the newest proven relevant height
<code>minAttestors</code>	the lowest attesor count registered for the source chain at the moment any of this agent's facts was admitted
<code>truncated</code>	the reviewer set exceeded the iteration cap, so the answer is incomplete and consumers must refuse it

Two field names are routinely misread, so we state them plainly. `minAttestors` is the network context at admission, not the set of signers behind a particular proof. `coveredThrough` is the newest proven fact, not a certificate that every block up to the tip was examined.

6. Turning facts into a price

`AgentHireEscrow` is an example consumer, not the product. It reads `facts()` under the hirer's own thresholds and computes a premium, which is the agent's cost of credit and is paid to the agent's owner.

```
coverage    = min(10000, breadthGrounded * 10000 / k)
cloneFactor = c * 10000 / (c + cloneDensityLB)
risk        = 10000 - coverage * cloneFactor / 10000
premium     = baseBps + (maxBps - baseBps) * risk / 10000
```

The premium is computed, never configured. A live example: agent 50724 carries a risk of 7,223 basis points, which yields a premium of 1,472 basis points, or 14.72%, sitting between the 1% floor and the 20% ceiling.

A record with holes is not priced at all. If `gapCount` exceeds zero the escrow reverts with `Gated`. Missing evidence is a gate, never a price, because otherwise an agent could buy its way past its own silence. The escrow can also refuse a `truncated` answer, facts admitted under a thin attesor set (`ThinQuorum`), and facts whose newest proven height is too far behind the attested tip (`Stale`).

7. Paying for evidence, in either direction

`CoverageBounty` lets anyone put money on a question. `fund` snapshots a four-part decision vector under the funder's thresholds: are there enough verified reviewers, are there no holes, are look-alikes under the limit, and is there any negative review. `proveAndClaim` admits new proofs and pays the full bounty only if one of those four predicates changed **within that call**.

Evidence that damages an agent pays exactly as much as evidence that helps it. Proof that changes nothing pays nothing and costs only gas. This is what makes the bureau self-filling without a subsidy: the people who care about an answer fund the evidence that settles it.

8. The scout

GroundedScout is an autonomous agent that decides what is worth proving. It logs four decisions on every cycle.

- **R1 targeting.** Open bounties first, then agents that were requested, then the most reviewed agents of the last seven days.
- **R2 evidence both ways.** Complete records of senior reviewers, and against the agent: negative reviews, revocations, reviewers who own agents, clone siblings, and the highest review index.
- **R3 timing and cost.** Skip what is already admitted; prove only when a bounty covers the gas; batch four proofs per transaction.
- **R4 acting.** Hire through the escrow when the facts clear its own thresholds, otherwise fund a bounty.

Anyone may run one. The scout has no privilege: it carries proofs and nothing else. A dishonest scout cannot insert a review that never happened, because the precompile rejects anything not included in an attested block and logs count only from the official registries. All it can do is **withhold**, and the registry's own per-pair index counter exposes that as `gapCount`. Contributions change completeness; they never change correctness.

9. Verifiability

`recomputeFromChain()` reads every `TxAdmitted` event from the bureau, locates each source transaction on Ethereum by block and index, fetches its proof again from the Attestcoin prover, and replays it through an off-chain copy of the contract logic. On 13 September 2026 it replayed all 33 admitted transactions and produced facts identical to `facts()` for agents 22771, 21548, 50283 and 50286. The MCP tool `tinjau_verify` exposes the same check to any agent that can speak the Model Context Protocol.

10. Reading claims, outside the fact path

Review documents assert payments in marketplace-specific formats that no schema covers. A Gemini model reads them and proposes `(network, txHash)` pairs, falling to the next model in a ladder when a quota runs out. Deterministic checks then decide: the hash must appear in the document, the chain must be readable by Attestcoin, and the prover together with the precompile must confirm inclusion. On agent 50283 the reader found six claimed payment transactions and **none** of them exists on the chain it names.

That output is a report. It is never written on chain, and no price depends on it.

11. Results on Creditcoin CC3 testnet

Contract	Address
<code>GroundedFacts</code>	<code>0x67394eC13E911ab0D3A26132BECa404F26e17a98</code>
<code>AgentHireEscrow</code>	<code>0xF801a8a01E018f3Bf9a648F4C095a53979282EEA</code>
<code>CoverageBounty</code>	<code>0xa27f14CD50BF334E7Fb09601cEf203745aADF569</code>

All three are verified on Blockscout. 33 Ethereum mainnet source transactions are admitted, the oldest at block 14,306,215 from 2 March 2022. The scout keeps adding, so that figure is a floor and not a ceiling. The contract suite carries 49 tests.

Worked outcomes

- **Agent 22771.** Three reviewers, each active between 97 days and four years before their first review, no gaps, no look-alikes, four attestors at admission. Premium 100 basis points. Hired.
- **Agent 21548.** Bounty #0 of 0.05 tCTC claimed by the scout with the batch that moved the decision, then hired at 100 basis points.

- **Agent 50283.** One reviewer who owns agents, review #97 proven with #1 to #96 unproven, nine clone siblings. Quote 2,000 basis points and `hire` reverts `Gated(1)`. No threshold opens it.

Precompile cost

Source height (Ethereum)	Continuity roots	verify gas
14,306,215 (2 Mar 2022, oldest admitted)	786	506,986
22,841,017	984	631,434
24,365,879 (agent 22771 registration)	122	127,746
25,489,494	7	62,292

Cost tracks the number of continuity roots rather than age, at roughly 55,000 gas plus 580 per root. A complete admitted proof, including decoding and storage, costs 0.3 to 0.6 million gas; a 52 KB mass-registration transaction carrying ten `Registered` logs costs 2,999,199.

Mainnet readiness

Checked read-only on CC3 mainnet: the proof for agent 22771's registration, fetched from the mainnet proof builder, returns `verify = true` from the `0x0FD2` precompile at the same 127,746 gas. The contract code, the registries and the proofs are unchanged; only the `chainKey` in the constructor differs. Mainnet `AttestorStash` reports seven attestors for Ethereum with a minimum bond of zero.

12. Trust assumptions

- Attestcoin moves trust away from RPC providers and indexers and onto Creditcoin's bonded attestor set. It does not remove trust. On mainnet the minimum bond is currently zero, and a consumer that cares can refuse facts admitted under a thin set.
- The precompile is trusted to verify inclusion correctly. Tinjau does not re-implement it.
- The registry addresses are constructor arguments with no admin, so they are fixed at deployment and visible to anyone reading the contract.
- An agent's own registration document is the owner's unverified claim. The interface labels it as such and never mixes it with proven facts.

13. Limits, stated up front

- **Ethereum mainnet only.** Sepolia is excluded by design, because a registry entry there costs nothing to mint and admitting it would let anyone fabricate a record for free. Base, where most ERC-8004 activity now lives, is not readable by Attestcoin today.
- **Admitted evidence with known gaps, never a complete history.** Holes below the highest proven index are visible; a newer review that nobody has proven is not.
- **Clone density is a lower bound.** Aged wallets can be bought, and honest multi-agent operators look like clone farms. The contract reports the count and draws no conclusion.
- **No independent consumer contract on Creditcoin yet.** The escrow is a worked example of one.
- **Evidence Exchange is partly live.** The decision-change payout and source-order conflict resolution work today. Batch claim, an adjudication receipt recording the decision before and after with the predicate that flipped, and hire re-pricing inside the same transaction are not built.

14. What comes next

1. Finish Evidence Exchange: batch claim, the adjudication receipt, and same-transaction re-pricing.

2. Consumer contracts beyond our own escrow. Any marketplace, lender or agent can already read `facts()` through `IAgentFacts`.
3. Additional source chains as Attestcoin reaches them. The registry table is a constructor argument, so no rewrite is required.
4. Distribution through the Model Context Protocol, so an agent consults the bureau before it hires rather than after it has paid.

Who pays. The hirer, as a premium on the job, and the premium goes to the agent's owner. That is deliberate: it makes a clean, fully proven record an asset its owner has a reason to protect.

Every figure in this document is either an on-chain measurement or a command anyone can rerun. The integration summary lists every testnet transaction with its gas at `ATTESTCOIN_INTEGRATION.md`; the evaluation dossier labels each claim as fact or inference at `docs/evaluation-dossier.md`. Live at tinjau.xyz, source at github.com/scientivan/Tinjau.